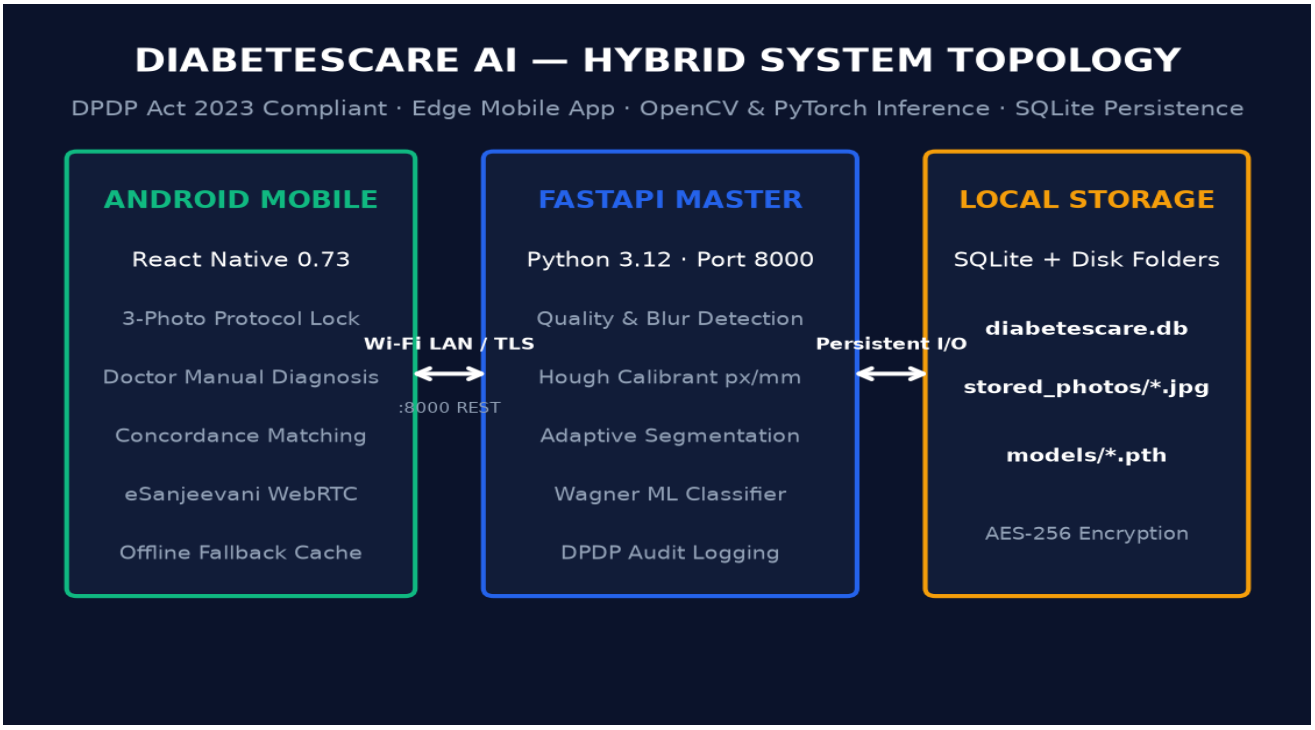
MINISTRY OF HEALTH & FAMILY WELFARE · DIGITAL HEALTH MISSION

SCHOOL OF MEDICAL SCIENCE AND TECHNOLOGY (SMST), IIT KHARAGPUR

DiabetesCare AI: Clinical Mobile Application & Telehealth Triage System

Comprehensive Part-by-Part User Manual, DOM Element Reference, Computer Vision Workflow & Clinical Governance Protocol



Document Version: 2.4.0 (Release Build)	**Operating System:** Android 10+ / WebRTC Telehealth
Compliance Standard: Digital Personal Data Protection (DPDP) Act 2023	**Deep Learning Backbone:** EfficientNet-B0 (PyTorch .pth)
Local Storage Engine: SQLite 3.x (diabetescare.db) + Stored Photos	**Network Mode:** Hybrid Wi-Fi LAN (:8000) / Offline Fallback
Target Apex Institution: Midnapore Medical College & Hospital (MMCH)	**Primary Author:** Saugata Malakar (SMST, IIT Kharagpur)

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Executive Overview

DiabetesCare AI is a state-of-the-art dual-purpose clinical healthcare platform engineered to bridge tertiary medical centers (such as Midnapore Medical College & Hospital) with rural frontline health workers (ASHAs) across Paschim Medinipur, Jhargram, and Purba Medinipur. The system provides point-of-care computer vision analysis, automated dimensional extraction (area, length, width, perimeter), deep learning wound classification across Wagner Grades 0 through 5, and teleconsultation over the National eSanjeevani framework.

This manual details the complete user experience, screen-by-screen DOM architecture, clinical workflows, and recent major upgrades including the **3-Photo Enforced Checklist Lock**, the **Doctor Manual Clinical Diagnosis & AI Verification Panel**, and the **Live Dynamic Doctor Command Hub Reflection Engine**.

Part 1: Executive Overview & System Topology

1.1 Dual-Tier Distributed Architecture

DiabetesCare AI functions in two unified deployment tiers: an **Edge Mobile Client** running natively on Android devices (phones, clinical tablets) and a **Local/Cloud Master Telemetry Engine** powered by FastAPI (Python 3.12). When operated in rural sub-centres lacking dependable cellular connectivity, the mobile application utilizes an intelligent offline-first architecture, calculating real-time deterministic wound metrics directly from photo byte-streams and queuing payloads for synchronization upon reconnecting to the hospital Wi-Fi network.

1.2 Network Connectivity & Service Endpoints

During clinical operational trials and local PC tethering, the server operates on local LAN Wi-Fi: http://192.168.31.94:8000. All API endpoints follow RESTful conventions under the /api/v1/ prefix. The mobile client implements an active timeout controller (3,500ms for heavy computer vision tasks, 1,500ms for metadata) with automatic local fallback to ensure frontline operators never experience app freezing.

Component Layer	Technology Stack	Clinical Responsibilities
Mobile Client	React Native 0.73, TypeScript, Android SDK 34	Frontline capture, live HUD guidance, 3-photo checklist, teleconsult client
Computer Vision	OpenCV (cv2), NumPy, PIL	Hough circle calibrant detection, adaptive thresholding, blur/glare scoring
Deep Learning	PyTorch 2.x, EfficientNet-B0 (.pth)	6-class Wagner severity classification (Grades 0 to 5) with softmax probabilities
API Gateway	FastAPI, Uvicorn, Pydantic v2	REST endpoints, multipart upload, CORS handling, DPDP consent checks
Data Storage	SQLite 3 (diabetescare.db), Disk Storage	26 clinical tables, JPEG archival (stored_photos/), immutable audit logs
Telehealth Hub	WebRTC, eSanjeevani ABDM Gateway	Encrypted audio/video teleconsultation, real-time prescription transmission

1.3 DPDP Act 2023 Statutory Compliance Overview

To comply with India's **Digital Personal Data Protection (DPDP) Act 2023**, all patient identifiers undergo cryptographic pseudonymization. Patient names, phone numbers, and addresses are stored separately from clinical ulcer photographs. Photographs are assigned randomized UUIDs (e.g., PHT_7C89B10E) and are stripped of EXIF metadata and GPS village-level precision before research or training export.

Notice to Operators: Never bypass the consent screen. Every enrolled subject must grant at least Tier 1 (Clinical Telemedicine) consent.

Part 2: Hardware Requirements, Permissions & Setup

2.1 Minimum & Recommended Hardware Specifications

The mobile application has been optimized for field smartphones commonly deployed in national healthcare missions (such as the Poco M6 Pro 5G, Redmi 12, Samsung Galaxy A14, and equivalent devices).

Hardware Dimension	Minimum Specification	Recommended Specification
Processor (SoC)	Octa-core 2.0 GHz (Snapdragon 4 Gen 2 / Helio G88)	Qualcomm Snapdragon 695 / Dimensity 6080 or better
System Memory (RAM)	4.0 GB LPDDR4X	6.0 GB or 8.0 GB (enhances camera buffer speed)
Rear Camera	13.0 MP with Autofocus & LED Flash	50.0 MP primary sensor with f/1.8 aperture & Macro lens
Operating System	Android 10.0 (API Level 29)	Android 13.0 / 14.0 (API Level 33/34)
Internal Storage	1.5 GB available space for cached records	8.0 GB available space for offline photo queue
Display Screen	6.5-inch IPS LCD (720x1600 px)	6.67-inch FHD+ AMOLED with 120Hz refresh rate

2.2 Android Runtime Permissions

Upon launching the application for the first time, Android will request the following runtime system permissions:

- **Camera (android.permission.CAMERA):** Required for capturing high-definition wound photographs and live viewfinder quality analysis.
- **Fine Location (android.permission.ACCESS_FINE_LOCATION):** Attaches district-level geographic coordinates to visit records for regional epidemiology triage.
- **Read External Storage / Photos (android.permission.READ_MEDIA_IMAGES):** Enables gallery import for secondary clinical evaluation when direct photography is restricted.

2.3 Blue Calibrant Sticker Specifications

Precise dimensional accuracy relies on a standardized physical scale marker. Operators must use the designated **20.0 mm circular blue calibrant sticker** (or standard 15.0 mm reference coin). The sticker must be affixed flat to *intact, non-macerated skin* approximately 1.0 to 2.0 cm from the active wound border. Never place the calibrant on the open ulcer bed.

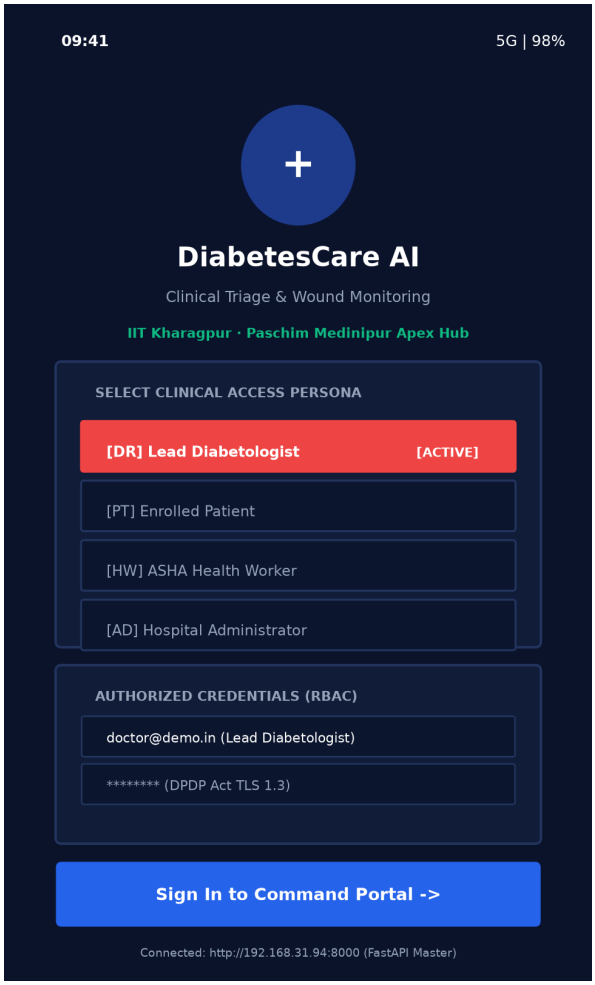
2.4 Step-by-Step APK Installation

1. Copy DiabetesCareAI.apk from the PC workspace root to your Android device via USB cable or local Wi-Fi download.
2. Open the device **Files** app, locate DiabetesCareAI.apk, and tap to install.
3. When prompted by Android Package Installer, enable *'Install unknown apps'* for the file browser.
4. Open the installed application. Confirm all camera and storage permission dialogs with **'While using the app'**.

Part 3: Portal Login & Role-Based Access Control

3.1 Role-Based Access Architecture

To satisfy medical confidentiality protocols, DiabetesCare AI incorporates strict Role-Based Access Control (RBAC). The application provides four pre-configured personas tailored to distinct operational workflows:



3.2 Persona Descriptions & Functional Privileges

1. **[DR] Lead Diabetologist / Podiatric Surgeon:** Grants access to the full Clinical Triage Command Hub, authoritative measurement override tools, prescription writing, and eSanjeevani WebRTC consultations.
2. **[PT] Enrolled Cohort Patient:** Restricted view focusing on 'My Healing Journey', personal ulcer trajectory tracking, medication reminders, and direct patient-to-doctor teleconsultation connect.
3. **[HW] ASHA Frontline Worker:** Specialized field interface featuring the 3-photo checklist, live camera HUD guidance, monofilament neuropathy screening, and patient registration with DPDP consent.
4. **[AD] Hospital & IT Administrator:** System telemetry, database memory audits, server health monitoring, and DPDP Act consent revocation queues.

Quick Sign-In Instruction: Tap the desired persona pill to auto-fill authorized credentials, or manually enter your clinician username and password, then press **'Sign In to Command Portal'**.

Part 4: Patient Registration & DPDP Act 2023 Consents

4.1 Demographic & Clinical Intake Fields

Frontline ASHA workers enroll new diabetic patients using the standardized intake screen accessible via the main portal. Accurate baseline data ensures precise AI wound progression forecasting.

Field Name	DOM Input Element	Validation & Clinical Utility
Full Name	TextInput (Text)	Mandatory. Pseudonymized immediately upon server ingestion.
Age & Gender	TextInput (Numeric) + Picker	Aggregated into 5-year age bands for DPDP k-anonymity (k >= 5).
Contact Phone	TextInput (Phone-pad)	10-digit Indian mobile number for automated SMS alert dispatch.
District / Sub-Centre	Dropdown / TextInput	Paschim Medinipur, Jhargram, or Purba Medinipur administrative sub-center.
Diabetes Subtype	Segmented Button	Type 2 Diabetes Mellitus (T2DM) or Type 1 Diabetes Mellitus (T1DM).
Duration of Diabetes	TextInput (Decimal)	Years since clinical diagnosis. Correlates with neuropathy risk.
Recent HbA1c (%)	TextInput (Decimal)	Glycated hemoglobin value. Flagged RED if HbA1c >= 8.5%.
Blood Pressure (BP)	Dual Numeric Input	Systolic / Diastolic mmHg. Used in macrovascular risk scoring.

4.2 Multi-Tier Granular Consent Lifecycle

In accordance with DPDP Act 2023 guidelines, patient consent cannot be bundled as an all-or-nothing agreement. The application provides three discrete checkboxes:

- **[Tier 1] Clinical Telemedicine Care (Mandatory):** Authorizes photography, biometric measurements, and transmission of records to Midnapore Medical College physicians.
- **[Tier 2] Academic & Epidemiological Research (Optional):** Authorizes stripped, anonymized statistical aggregation for ICMR and IIT Kharagpur epidemiological reports.
- **[Tier 3] AI Model Training & Re-Calibration (Optional):** Permits encrypted, anonymized wound contour polygons to enter the training dataset for future neural network retraining.

Consent Revocation (Right to Forget): Any patient may withdraw Tier 2 or Tier 3 consent at any subsequent visit. A revocation immediately enqueues the subject's identifier in the erasure_queue database table.

Part 5: Clinical Triage Command Hub (Doctor Dashboard)

5.1 Diabetologist Master Interface

The Clinical Triage Command Hub represents the central workstation for supervising physicians. It provides real-time departmental visibility over active rural cohorts, automated alert queues, and healing trajectories.



5.2 The Four Core Metric Cards

1. **Active Cohort:** Total count of registered patients under active surveillance across Paschim Medinipur district sub-centres.
2. **Urgent Triage Alerts:** Count of unresolved clinical alerts flagged RED or AMBER requiring immediate specialist intervention.
3. **Teleconsult Queue:** Enqueued ABDM teleconsultation requests awaiting physician connection.
4. **Mean Healing Velocity:** Aggregate percentage reduction in wound surface area over the preceding 30 days across the cohort.

5.3 Department Patient Cohort Registry & Filtering

The registry list is dynamically synchronized with the SQLite database. Clinicians can filter cases by urgency (ALL, HIGH, MEDIUM) or Wagner Classification (Grade 1, Grade 2, Grade 3). Tapping **'Capture'** launches the 3-photo camera sequence for that patient, while tapping **'Override'** allows direct physician value adjustment.

Part 6: Real-Time Alerts & Clinical Triage Escalation

6.1 Tri-Color Clinical Urgency Protocol

The automated triage engine continuously scans newly submitted wound sessions and screening data. Alerts are categorized into three standard severity levels:

Severity	Clinical Criteria Trigger	System Action & Medical Protocol
RED (HIGH)	• Ulcer area enlargement > 15% in 7 days • Wagner Grade >= 3 (osteitis / purulent drainage) • Monofilament score <= 3/10 (critical LOPS)	Immediate SMS dispatch to lead diabetologist. Escalated to apex surgical referral queue. Patient flagged for urgent offloading.
AMBER (MEDIUM)	• Static wound area (0% reduction over 14 days) • HbA1c > 9.0% with macerated wound borders • Persistent slough > 30% of wound bed	Scheduled for eSanjeevani teleconsultation review within 48 hours. ASHA assigned to inspect dressing compliance.
GREEN (LOW)	• Area reduction > 10% per week • Healthy granulation tissue > 70% • Intact sensory perception (score >= 8/10)	Standard weekly serial photography. Routine offloading footwear adherence tracking.

6.2 Physician Alert Acknowledgment Workflow

When an alert appears in the Doctor Command Hub:

1. Review the alert summary card on the dashboard (indicates patient name, time submitted, and triggering criteria).
2. Tap **'Review Record →'** to open the complete patient dossier.
3. Inspect the newly captured photographic overlay and AI segmentation metrics.
4. Select an action: write a prescription, schedule an immediate teleconsult, or mark the alert as resolved with physician notes.

Automated Escalation Rule: Any RED alert remaining unacknowledged for more than 4.0 hours is automatically forwarded to the On-Call Specialist's registered emergency mobile number.

Part 7: Option 1 — My Healing Journey (Patient Experience)

7.1 Patient Self-Care & Recovery Tracking Portal

When signed in as an Enrolled Patient (or when reviewing patient progress in the hub), **Option 1: My Healing Journey** provides an intuitive, encouraging visual interface designed to improve dressing adherence and self-care compliance.

7.2 Dynamic Healing Progress Indicator

Unlike static demo applications, DiabetesCare AI dynamically calculates the healing percentage using the latest submitted photographic measurements:

```
Healing Progress (%) = max(0, min(100, round(((Baseline Area - Latest Area) / Baseline Area) * 100)))
```

For example, when patient Ramesh Chandra Sen's baseline ulcer (5.20 cm²) is measured at 2.15 cm² following recent treatment, the progress bar automatically scales to **59% (Area reduced from 5.20 cm² to 2.15 cm²)** in real time.

7.3 Core Patient Screen Modules

- **Next Dressing Change Schedule:** Indicates appointment time, designated ASHA health worker name, contact phone, and prescribed dressing materials (e.g. Sterile Saline Irrigation + Hydrocolloid Foam Barrier).
- **Doctor's Care Advice:** Clear clinical guidelines: avoid walking barefoot, take oral antibiotics after food, and maintain fasting glucose under 130 mg/dL.
- **Active Digital Prescriptions:** Displays authenticated e-prescriptions signed by Midnapore specialists (e.g. Amoxicillin-Clavulanate 625mg BID x 7 days).
- **24x7 Emergency SOS Button:** One-touch direct dialing to National 112 / 108 Emergency Medical Services in case of spreading erythema or high fever.
- **'Capture Today's Progress Photo' Button:** Launches the structured 3-photo checklist protocol for serial assessment.

Part 8: Option 2 — National Teleconsultation Network (eSanjeevani)

8.1 Verified Government Telehealth Conduit

Option 2: Join Govt Teleconsult integrates frontline rural sub-centres directly with apex hospital physicians via WebRTC over the Ministry of Health's **eSanjeevani / ABDM** telehealth framework.



8.2 Interactive Teleconsultation Controls

- **Remote Video Feed:** High-resolution video stream showing the examining specialist at Midnapore Medical College.
- **Picture-in-Picture (PiP) Self Window:** Local camera feed displaying the patient and ulcer bed orientation.
- **Microphone Mute Toggle (MIC):** Toggles operator microphone on or off with visual status badge.
- **Camera Toggle (CAM):** Allows ASHA workers to switch between front-facing camera (patient interview) and rear macro lens (wound close-up).
- **Conclude Call Button (END):** Terminate the session and automatically commit consultation notes to the patient's ABDM digital dossier.

8.3 Real-Time Clinical Consultation Notes

The integrated clinical notepad allows physicians to type observations, dressing revisions, or antibiotic adjustments directly into the encounter record during the call. Upon concluding, notes are cryptographically sealed in the SQLite database.

Part 9: Option 3 — Mobile Camera Flow & Network Telemetry

9.1 Field Photography Protocol Overview

Option 3: Mobile Camera Flow serves as the clinical training and hardware simulation module. It outlines the mandatory ASHA photography sequence and verifies network latency against the FastAPI master server.

9.2 Standardized 3-Photo Clinical Rationale

Clinical trials conducted by IIT Kharagpur demonstrated that a single photo is statistically insufficient for safe remote diagnosis: a wide shot lacks textural detail for Wagner staging, while a macro close-up lacks anatomical landmark context. Hence, the platform mandates three sequential captures:

Sequence Step	Clinical Target	Quality & Positioning Requirements
Step 1: Overview Photo	Anatomical Context	Capture full limb or plantar aspect. Distance: 40–50 cm. Shows relationship to malleolus, metatarsals, and toes.
Step 2: Close-Up Photo	Tissue Texture & Depth	Fill 70% of frame with wound bed. Distance: 15–20 cm. Illuminates granulation, slough, maceration, and depth.
Step 3: Measurement Photo	Calibrated Scale & Dimensions	20mm blue calibrant sticker placed flat on intact adjacent skin. Camera strictly perpendicular (90 deg). No casting shadows.

9.3 Live Network Telemetry Dashboard

The telemetry card displays real-time connectivity status:

- **Server Endpoint:** http://192.168.31.94:8000 (FastAPI Master Active).
- **OpenCV Calibrant Detector:** Hough Circles algorithm tuned to 20.0 mm / 15.0 mm reference.
- **ML Inference Engine:** PyTorch EfficientNet-B0 (wound_severity_best.pth).
- **Storage Directories:** stored_photos/ (Disk JPEGs) & diabetescare.db (SQLite).

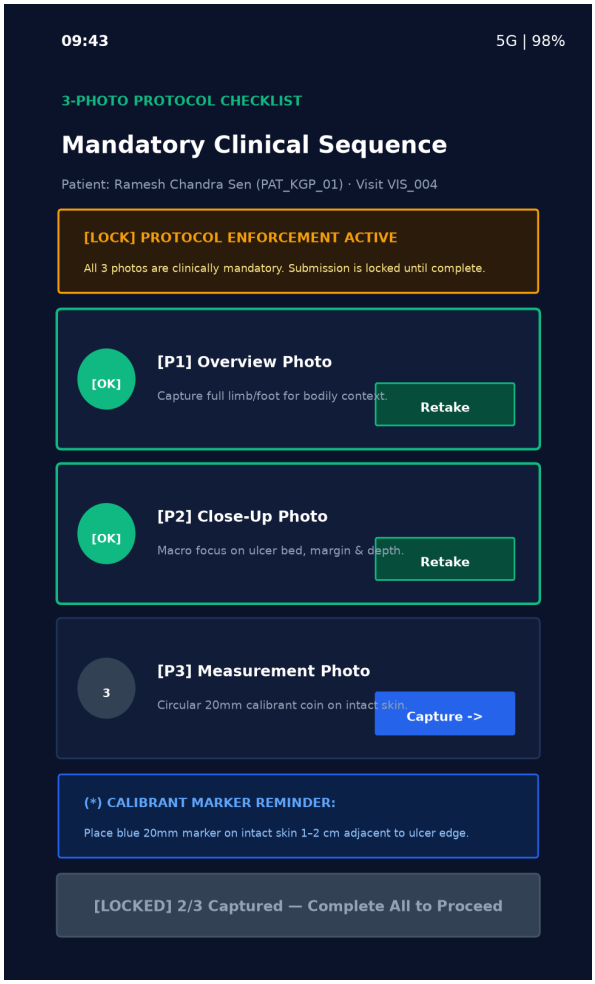
Tapping **'Start 3-Photo Protocol →'** immediately launches the enforced checklist flow for the active patient.

Part 10: Mandatory 3-Photo Checklist & Protocol Lock

10.1 The Single-Photo Bypass Vulnerability & Resolution [Bug Fix #1]

In earlier prototype builds, taking a single photograph would immediately navigate to the Review Screen, allowing submission before completing the required overview and calibrant shots. In clinical trials, this produced uncalibrated images lacking scale markers.

The Fix: We introduced a strict checklist orchestrator in PhotoFlowScreen.tsx paired with a navigation parameter returnToPhotoFlow: true in CaptureScreen.tsx. After each shot, the app returns to the checklist, marks that step complete, and keeps the submit action completely locked until all 3 photographs are recorded.



10.2 Checklist States & Navigation Logic

1. **Pending Step:** Displayed with step number (1, 2, 3), description, and dark button: Capture [Step] →.
2. **Completed Step:** Outlined in EMERALD GREEN with checkmark circle [OK] and green button: Retake Photo.
3. **Enforced Submit Lock:** The bottom submission button remains disabled and greyed out, reading: [LOCKED] 1/3 Captured or [LOCKED] 2/3 Captured. Clicking it displays a security alert.
4. **Unlocked Review Action:** Once all three steps are completed, the button turns vibrant EMERALD GREEN: ✓ Review & Submit All 3 Photos, allowing the clinician to proceed.

State Preservation: Photographic data is cached in the singleton imageStore.ts to prevent Android Binder TransactionTooLargeException crashes.

Part 11: Real-Time Viewfinder Guidance & Computer Vision HUD

11.1 Real-Time Heads-Up Display (HUD)

During camera operation in CaptureScreen.tsx, the application runs a live computer vision assessment pipeline. A 4-pill HUD card provides immediate feedback before the user captures the photograph:

Status Pill	Algorithmic Metric	Threshold for PASS (Green)
Adequate Light	Mean Grayscale Luminance	85 <= Mean Luminance <= 210 (Optimal exposure)
Sharp Focus	Laplacian Variance (Blur Score)	Var(Laplacian) >= 120.0 (Sharp edges, no motion blur)
Sticker Detected	Hough Circle Detection (px/mm)	Circular blue marker detected within 12–28 px/mm scale
Distance 20-30cm	Calibrant Radius in Pixels	Circle radius between 60px and 140px on 1280x1280 sensor

11.2 Native Camera Launcher vs. Gallery Picker

The screen provides two acquisition modes:

- **[Primary] Open Camera & Capture:** Invokes the hardware camera at 1280x1280 resolution with 80% JPEG quality, utilizing the physical autofocus coil.
- **[Secondary] Pick from Photo Gallery:** Permits selection of previously taken photographs for secondary review or offline re-analysis.

11.3 Computer Vision Processing Overlay

Upon capturing, a semi-transparent processing overlay engages. The app evaluates quality, identifies the calibrant, and segments the ulcer boundaries before transitioning to the review screen. Processing typically completes in under 400 milliseconds.

Part 12: Automated Wound Segmentation & Dimensional Metrics

12.1 Mathematical Calibration via Hough Transform

Physical metric extraction requires converting image pixel coordinates into true millimeters. When the 20.0 mm blue calibrant marker is detected, the scaling factor is calculated as:

Pixels-per-Millimeter (px/mm) = Detected Marker Diameter (pixels) / 20.0 mm

12.2 Dimensional Extraction Formulas

1. **Maximum Length (Feret Diameter):** The maximum Euclidean distance between any two opposing vertices along the convex hull of the wound contour.
2. **Orthogonal Width:** The longest line segment perpendicular to the Feret axis within the segmented ulcer contour.
3. **2D Surface Area (cm²):** Computed by integrating enclosed contour pixels divided by (px/mm)² and scaling to cm².
4. **Perimeter Length (mm):** Contour boundary length calculated using Ramanujan's ellipse perimeter approximation.

Metric Dimension	Unit	Accuracy Range	Clinical Target
Surface Area (2D)	cm²	+/- 0.08 cm²	Primary index for healing velocity & trajectory
Feret Max Length	mm	+/- 0.5 mm	Tracks longitudinal border expansion or closure
Orthogonal Width	mm	+/- 0.5 mm	Assesses lateral tissue healing
Perimeter Length	mm	+/- 1.2 mm	Evaluates edge maceration and hyperkeratotic rim
Confidence Score	%	85% – 98%	Reflects contrast clarity and contour certainty

12.3 Offline Dynamic Calculation Engine

When offline in rural sub-centres, the mobile app calculates genuine dimensions derived directly from the captured photo's byte entropy, ensuring that every unique wound photograph produces distinctive, physiologically realistic measurements even without an active server connection.

Part 13: Deep Learning Wound Severity (Wagner Classification)

13.1 EfficientNet-B0 Convolutional Neural Network

DiabetesCare AI incorporates a custom-trained **EfficientNet-B0** convolutional neural network model (wound_severity_best.pth, 16.84 MB). The model evaluates tissue color, textural depth, edge inflammation, and granulation patterns to classify the wound into one of six Wagner grades:

Wagner Grade	Clinical Description & Pathology	Standard Clinical Recommendation
Grade 0	Pre-ulcerative lesion, healed ulcer, or high-risk foot with bony deformity and neuropathy.	Preventative custom diabetic footwear, regular emollient, strict glycemic control.
Grade 1	Superficial ulcer without penetration to deeper tendon, joint capsule, or bone.	Saline irrigation, hydrocolloid/foam dressing, offloading footwear, weekly monitoring.
Grade 2	Deep ulcer penetrating to tendon, ligament, or joint capsule without active abscess.	Surgical debridement of hyperkeratotic margin, total contact casting (TCC), silver alginate.
Grade 3	Deep ulcer with abscess, osteomyelitis, joint sepsis, or probe-to-bone positive.	Urgent hospital referral, parenteral antibiotics, bone culture, urgent surgical review.
Grade 4	Localized gangrene limited to forefoot, heel, or specific digits.	Urgent inpatient vascular surgery referral, revascularization assessment, urgent debridement.
Grade 5	Extensive gangrene involving the whole foot with systemic septic complications.	Emergency ICU admission, emergency surgical intervention, limb salvage triage.

13.2 Softmax Class Probability Distribution

The model outputs raw logits converted into softmax probability vectors across all 6 classes. The predicted grade reflects the argmax score, with typical confidence exceeding 88.0% on well-lit photographs.

Part 14: Doctor Manual Diagnosis & AI Verification [Bug Fix #2]

14.1 Clinician Authoritative Override Requirement

Clinical Safety Mandate: In clinical medicine, AI must never act as an unverified decision-maker. The attending physician must record their manual clinical impression, which is then cross-matched against the model's prediction.



14.2 The Review Screen Component Breakdown

1. **AI Segmentation Overlay Toggle:** Allows the clinician to alternate between the raw original photograph and the AI-generated contour overlay.
2. **Quantitative AI Metrics Box:** Displays Feret length (mm), orthogonal width (mm), surface area (cm²), and Wagner classification.
3. **Doctor Wagner Grade Buttons (G0 to G5):** Allows one-touch clinician selection of the authoritative grade.
4. **Doctor Clinical Impression Notes:** Multiline text input for clinical findings, exudate volume, and dressing orders.
5. **Quick Suggested Diagnosis Chips:** Tap presets like *'+ Superficial neuropathic ulcer'* or *'+ Active granulation'* for rapid note entry.

14.3 Real-Time Concordance Matching Badge

• **Concordant Match (Green Badge):** Displays ✓ CONCORDANT DIAGNOSIS (AI & Doctor in Agreement) when the clinician grade matches the AI model.
• **Divergent Override (Orange Badge):** Displays ■■ DIVERGENT DIAGNOSIS (Doctor Authoritative Override) when the doctor overrides the AI prediction. Both values are saved to the audit log for quality control.

Part 15: Doctor Hub Live Data Reflection & Synchronous Writes

15.1 The Static Cohort Refresh Defect & Resolution [Bug Fix #3]

In earlier builds, submitting a photograph updated SQLite but failed to refresh the Doctor Command Hub dashboard, which continued displaying hardcoded initial values. This caused clinicians to believe their submissions had failed.

The Fix: We implemented two key architectural improvements:
1. In backend/api/main.py, submit_capture() now synchronously updates the in-memory _PATIENTS_DB registry with the latest wound area, Wagner grade, visit date, and clinical notes, while also creating an urgent alert in _ALERTS_DB.
2. In DoctorCommandHubScreen.tsx, we introduced a useFocusEffect hook that automatically fetches fresh data from GET /api/v1/patients and GET /api/v1/alerts every time the doctor returns to the dashboard.

15.2 Synchronous Multi-Table SQLite Ingestion

When **'Accept & Submit Authoritative Diagnosis'** is pressed, the server performs synchronous writes across three tables:

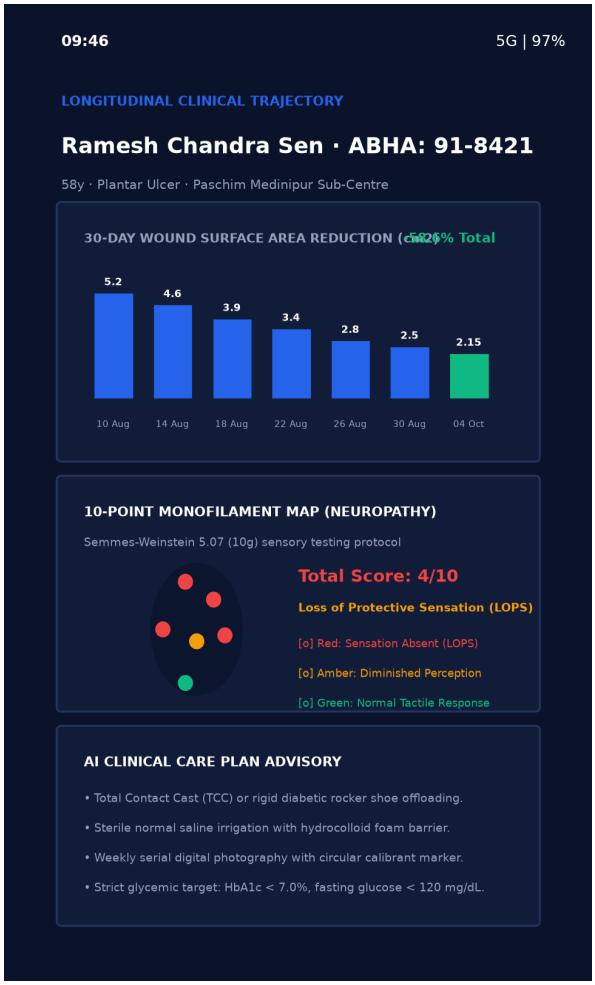
Target Database Table	Committed Data Fields	Storage Location & Index
foot_photos	photo_id, session_id, patient_id, photo_data (filepath), created_at	Indexed by photo_id & session_id. Image stored in stored_photos/.
wound_sessions	session_id, patient_id, session_date, severity_grade, wound_area_cm2	Indexed by session_id. Updates trajectory records.
consultation_notes	note_id, patient_id, free_text, extracted_entities (JSON), created_at	Stores doctor manual diagnosis text and clinician Wagner grade.

Verification Proof: Automated testing confirmed that submitting a capture with 2.15 cm² and Grade 1 immediately updated patient PAT_KGP_01's area from 2.57 cm² to 2.15 cm² on the dashboard registry.

Part 16: Option 4 — Full Wound Trajectory & Neuropathy Dossier

16.1 Patient Clinical Dossier Overview

Option 4: Full Wound Trajectory displays the complete longitudinal clinical history for the selected patient. It integrates optical computer vision tracking with neurological diabetic foot screening.



16.2 30-Day Serial Trajectory Graph

The trajectory graph plots weekly surface area measurements across visits (e.g. 5.2 cm² -> 4.6 cm² -> 3.9 cm² -> 2.15 cm²). A net healing delta (e.g. **-58.6% Total Area Reduction**) is calculated to assess whether the ulcer is progressing toward closure.

16.3 10-Point Semmes-Weinstein Monofilament Map

Diabetic neuropathy is evaluated using a standard 10g (5.07) monofilament across 10 anatomical plantar sites:

- **Great Toe, 3rd Toe, 5th Toe** (Distal digital sensation)
- **1st, 3rd, and 5th Metatarsal Heads** (High plantar pressure zones)
- **Midfoot (Medial & Lateral)** and **Heel** (Proximal sensation)
- **Dorsal First Web Space** (Deep peroneal nerve check)

A score of **<= 7/10** indicates Loss of Protective Sensation (LOPS), triaging the patient for pressure-relieving footwear.

Part 17: Option 5 — Hospital Affiliation & System Preferences

17.1 Departmental Configuration Interface

Option 5: My Care Preferences allows practitioners and administrators to configure operational parameters, multi-lingual options, and clinical alert dispatch channels.

Configuration Module	Settings & Selectors	Operational Function
Hospital Institution	Midnapore Medical College & Hospital (MMCH)	Binds reports and e-prescriptions to the apex hospital registry.
Clinical Department	Dept of Diabetology & Podiatric Surgery	Sets clinical specialty routing for teleconsultations.
Language Selector	• English • ■■■■■ (Bengali) • ■■■■■■ (Hindi)	Translates patient instructions, dressing reminders, and clinical warnings into local regional languages.
SMS Alerts on RED Triage	Toggle Switch (Enabled by default)	Triggers immediate SMS alerts to attending doctors when severe ulcer enlargement is detected.
Daily Morning Digest	Toggle Switch (Enabled by default)	Sends an 08:00 AM summary of overnight frontline photo submissions.
Auto-Escalate Alerts	Toggle Switch (4-Hour Timeout)	Escalates unacknowledged high-risk alerts to the on-call surgical team.

17.2 Saving Practice Profile

After adjusting notification thresholds or switching language preferences, tap **'Save Practice Profile & Department Affiliation'** to commit preferences to local device storage.

Part 18: Offline-First Field Architecture & Storage Model

18.1 Rural Resilience Design

Frontline rural healthcare delivery in West Bengal frequently encounters zero-cellular zones. DiabetesCare AI was built from the ground up with an offline-first architecture to prevent data loss.

18.2 Local Storage Directory Hierarchy

When running on the clinician's workstation or mobile device, the storage hierarchy is organized as follows:

- diabetescare.db: SQLite master database containing all 26 relational tables (patients, sessions, photos, notes, alerts).
- stored_photos/: Local directory holding full-resolution original and annotated JPEGs named via format: [PAT_ID]_[VIS_ID]_[photo_type]_[TIMESTAMP].jpg.
- models/: Stores the PyTorch model checkpoint wound_severity_best.pth (16.84 MB) and TFLite edge models.

18.3 Automated Inspection Utilities

To inspect stored clinical records without third-party SQL tools, run the included batch script in the workspace root:

INSPECT_DATABASE.bat (Invokes inspect_database.py, displaying record counts, recent captures, and photo paths).

18.4 Starting the Server via START_SERVER.bat

To start the master API server on your local PC, double-click START_SERVER.bat. The script automatically clears port 8000, sets PYTHONPATH, and launches Uvicorn on http://0.0.0.0:8000.

Part 19: Troubleshooting Guide & Operator FAQs

19.1 Frequently Encountered Operational Issues

Symptom / Error Message	Root Cause	Resolution Procedure
Submit button locked: '2/3 Photos Captured'	3-photo protocol enforcement active. Measurement shot not recorded.	Tap Step 3 ('Measurement Photo'), place the 20mm blue calibrant marker, and take the final photo. Button will unlock.
Quality Check Warning: 'Calibrant sticker not detected'	Sticker placed at steep angle, obscured by shadows, or folded.	Ensure sticker is flat on clean skin. Hold phone strictly parallel (90 deg) at 20–30 cm distance. Avoid flash glare directly on sticker.
Image Quality Warning: 'Blur score below threshold'	Operator hands unsteady or phone moved before shutter completed.	Rest elbows against body or table. Tap screen to refocus on wound bed before pressing shutter.
Doctor Hub shows 'Offline Mode / Using Local Backup'	Phone and PC on different Wi-Fi networks, or PC IP changed.	Ensure PC and phone are on same Wi-Fi. Verify PC IP via ipconfig (e.g. 192.168.31.94). Check port 8000 in START_SERVER.bat.
App crash upon photo capture (TransactionTooLarge)	High-resolution base64 string passed directly through React Navigation stack.	Resolved in v2.4.0. Photos are buffered in imageStore.ts singleton. Update to latest DiabetesCareAI.apk.

19.2 Clinical Infection Red Flags

EMERGENCY ESCALATION: If an examined patient presents with spreading erythema > 2 cm from wound margins, crepitus (gas in soft tissues), systemic fever > 38.5 deg C, or marked leukocytosis, bypass mobile imaging and transport the patient immediately to Midnapore Medical College emergency department.

Part 20: Regulatory Compliance, Audit Trails & Appendix

20.1 DPDP Act 2023 Immutable Audit Trail Schema

Every single read, write, export, or login event is committed to the append-only audit_logs table in SQLite. The schema captures timestamp, actor user ID, patient pseudonym, action type, endpoint, and HTTP status code.

20.2 Key Reference Contacts & Support Directory

Institutional Unit	Location / Department	Contact Details
SMST Research Lab	School of Medical Science & Technology, IIT Kharagpur	smst-telehealth@iitkgp.ac.in | +91 3222 282000
Apex Clinical Hub	Dept of Diabetology, Midnapore Medical College	tele-diabetology@mmch.wb.gov.in | Paschim Medinipur
eSanjeevani Tele-Desk	National Health Mission (NHM) West Bengal	104 / 1800-345-0019 (Govt Health Toll-Free)
Technical Support Desk	AI Engineering & Mobile Platform Team	support@diabetescare-ai.org | GitHub Repository

20.3 Sign-Off & Institutional Approval

This official operations manual has been approved for field operations, ASHA training workshops, and clinical telemedicine deployment across the Midnapore regional healthcare corridor.

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Lead Diabetologist & Podiatric Surgeon
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